

Cursor Debugging Practice

The Case of the Bug Bounty Arcade

MISSION BRIEF

The Bug Bounty Arcade is malfunctioning just before alien visitors arrive. You will use Cursor, browser evidence, and a repeatable debugging process to solve five cases. The goal is not to guess. The goal is to observe, explain, make one small change, and prove the fix works.

Recommended grade	6-12
Time	55-75 minutes
Tools	Cursor + web browser
Files	index.html, styles.css, script.js

Detective motto: Bugs are clues, not disasters.

1. Learning Goals

- ☐ Describe a bug using Expected vs. Actual behavior.
- ☐ Reproduce a bug with exact steps.
- ☐ Use browser console messages and code clues as evidence.
- ☐ Give Cursor a strong debugging prompt instead of saying “fix it.”
- ☐ Apply one small, safe change at a time.
- ☐ Test the original case and nearby features after each fix.
- ☐ Record what changed and what was learned.

Important Rule
Do not accept a large rewrite from AI. Ask Cursor to explain the likely cause, suggest one small fix, and tell you how to test it.

The Detective Trail

1 Observe	2 Reproduce	3 Gather clues	4 Ask Cursor	5 Test	6 Record
What is wrong?	Can it happen again?	Console, file, line	Use a precise prompt	Did the fix work?	What changed?

2. Set Up the Investigation

1. Unzip the starter project

Download and unzip Bug_Bounty_Arcade_Student_Starter.zip. Keep the folder name simple and do not move individual files out of the folder.

2. Open the folder in Cursor

In Cursor, choose File > Open Folder, then select the student_app folder. Confirm that index.html, styles.css, and script.js appear in the Explorer.

3. Run the app

Open index.html in a browser. A local server is recommended, but opening the HTML file directly also works for this project.

4. Arrange your workspace

Keep Cursor on one side and the browser on the other. Open the browser Developer Tools and select the Console tab. On Windows, F12 usually opens Developer Tools.

5. Do not fix anything yet

Click every button once, submit the form once, enter ORBIT once, and resize the browser. Write observations before changing code.

Safety Check

Never paste passwords, API keys, private student information, or personal data into an AI prompt.

3. Your Bug Log

Complete one row before and after each fix. A clear record is part of the assignment.

Case	Expected	Actual	Steps / evidence	Likely cause	Fix tried	Test result
1						
2						
3						
4						
5						

Strong Cursor Prompt Template

Copy, complete, and paste this into Cursor Chat

I am debugging a beginner HTML/CSS/JavaScript web app.

When I [exact action], I expected [expected result]. Instead, [actual result] happened.

Steps to reproduce:

- 1. [step]
- 2. [step]
- 3. [step]

Evidence/error: [exact message or ‘no visible error’].

Relevant files: [file names].

Please explain the likely cause, point to the smallest relevant code area, suggest one safe fix, avoid unrelated rewrites, and give me a short testing checklist.

Case 1: The Hungry Alien

Mission

Click Feed Alien. The snack number and green meter should increase until they reach 5.

A. Observe and reproduce

What exact action did you take?

What did you expect?

What actually happened?

Can you repeat the same result?

What is the first useful Console message, if any?

B. Investigate without random changes

- ☐ Compare the button ID in index.html with the ID used in script.js.
- ☐ Look at the first red error in the browser Console.
- ☐ Ask Cursor what “Cannot read properties of null” means.

C. Ask Cursor with clues

Write or paste your completed prompt here:

D. Predict before applying the fix

I think the bug happens because: _____

The smallest change I plan to make is: _____

E. Test the fix

- ☐ Click once: count becomes 1 and meter becomes 20%.
- ☐ Click five times: count stops at 5.
- ☐ Other buttons still work.
- ☐ I checked the Console for new errors.
- ☐ I recorded the result in the Bug Log.

Case 2: The Score That Glues Numbers Together

Mission

Add 5 points, then add 10 points. The total should be 15, not a joined number.

A. Observe and reproduce

What exact action did you take?

What did you expect?

What actually happened?

Can you repeat the same result?

What is the first useful Console message, if any?

B. Investigate without random changes

- ☐ Use `typeof` or `console.log` to inspect the selected value.
- ☐ HTML form values are often text, even when they look like numbers.
- ☐ Ask Cursor how to safely convert a string to a number.

C. Ask Cursor with clues

Write or paste your completed prompt here:

D. Predict before applying the fix

I think the bug happens because: _____

The smallest change I plan to make is: _____

E. Test the fix

- ☐ Refresh the page.
- ☐ Add 5, then 10: total is 15.
- ☐ Add 20: total is 35.
- ☐ I checked the Console for new errors.
- ☐ I recorded the result in the Bug Log.

Case 3: The Disappearing Pilot

Mission

Register a pilot. The confirmation should remain visible and the page should not reload.

A. Observe and reproduce

What exact action did you take?

What did you expect?

What actually happened?

Can you repeat the same result?

What is the first useful Console message, if any?

B. Investigate without random changes

- ☐ Watch the browser carefully after Submit.
- ☐ The bug involves the default behavior of an HTML form.
- ☐ Search the code for the submit event.

C. Ask Cursor with clues

Write or paste your completed prompt here:

D. Predict before applying the fix

I think the bug happens because: _____

The smallest change I plan to make is: _____

E. Test the fix

- ☐ Enter Captain Nova and submit.
- ☐ The welcome message remains visible.
- ☐ The browser does not refresh.
- ☐ I checked the Console for new errors.
- ☐ I recorded the result in the Bug Log.

Case 4: The Invisible Secret Message

Mission

Enter ORBIT and click Check Code. A success message should appear below the button.

A. Observe and reproduce

What exact action did you take?

What did you expect?

What actually happened?

Can you repeat the same result?

What is the first useful Console message, if any?

B. Investigate without random changes

- ☐ Compare the message element ID in HTML and JavaScript.
- ☐ A correct value can still fail if JavaScript targets the wrong element.
- ☐ Use the Console error as evidence.

C. Ask Cursor with clues

Write or paste your completed prompt here:

D. Predict before applying the fix

I think the bug happens because: _____

The smallest change I plan to make is: _____

E. Test the fix

- ☐ ORBIT shows “Access granted!”
- ☐ A wrong code shows “Try again.”
- ☐ No Console error appears.
- ☐ I checked the Console for new errors.
- ☐ I recorded the result in the Bug Log.

Case 5: The Sideways Phone Screen

Mission

Make the browser narrow. The page should fit the screen without horizontal scrolling.

A. Observe and reproduce

What exact action did you take?

What did you expect?

What actually happened?

Can you repeat the same result?

What is the first useful Console message, if any?

B. Investigate without random changes

- ☐ This is a CSS issue, not JavaScript.
- ☐ Inspect the .case-grid rule.
- ☐ Fixed pixel widths can be larger than a phone screen.

C. Ask Cursor with clues

Write or paste your completed prompt here:

D. Predict before applying the fix

I think the bug happens because: _____

The smallest change I plan to make is: _____

E. Test the fix

- ☐ At about 375px wide, there is no horizontal scrollbar.
- ☐ Cards remain readable.
- ☐ Desktop layout still looks good.
- ☐ I checked the Console for new errors.
- ☐ I recorded the result in the Bug Log.

4. Final Challenge: Add a Helpful Empty-Input Warning

Bonus Mission

After the five bugs are fixed, improve the Pilot Registration form so a blank or spaces-only name cannot be submitted. Show a friendly message such as “Please enter a pilot name.”

- ☐ First describe the current behavior and desired behavior.
- ☐ Ask Cursor for a small validation change only.
- ☐ Use `.trim()` so spaces do not count as a name.
- ☐ Test blank input, spaces-only input, and a real name.
- ☐ Make sure the page still does not reload.

5. Detective Debrief

Which bug was easiest to describe? Why?

Which clue was most useful: visible behavior, Console error, or code comparison?

What did Cursor explain that you did not understand before?

How did testing prevent you from accepting a bad fix?

What will you do first the next time an app does not work?

6. Completion Checklist and Scoring

Category	Evidence	Points
Observation	Expected and Actual are specific for each case	5
Reproduction	Steps can be followed by another student	5
AI use	Prompts include clues and request a small fix	5
Fixes	Five cases attempted; changes are focused	10
Testing	Original and nearby behaviors are checked	10
Reflection	Debrief explains learning honestly	5

Success Standard
A student may earn strong credit even if one bug is not fully fixed, as long as the observation, evidence, attempted fix, test result, and next step are clearly documented.

Teacher Notes and Answer Key

Keep this page separate from students during the investigation.

Case	Root cause	Minimal fix	Concept
1	HTML uses feedAlienBtn; JavaScript searches for feedBtn.	Change feedBtn to feedAlienBtn.	DOM lookup and ID matching
2	select.value is a string, causing concatenation.	Wrap the value in Number(...).	Data types and conversion
3	Form submission reloads the page.	Call event.preventDefault().	Browser default behavior
4	JavaScript searches for caseMessage; HTML uses statusMessage.	Change caseMessage to statusMessage.	DOM lookup and null errors
5	.case-grid has a fixed 1100px width.	Use width: min(1100px, 100%) or max-width with width: 100%.	Responsive CSS

Suggested pacing

- 5 min: setup and observation
- 8 min: teacher models one prompt without applying the fix
- 30-40 min: student cases
- 10 min: peer test and bug-log review
- 5 min: debrief

Teaching reminders

- Ask students to say “I expected X, but Y happened.”
- Require one small change at a time.
- Do not let students paste an entire project before they can describe the bug.
- Have partners test each other’s fixes.
- Praise accurate documentation even when a bug takes more than one attempt.